CHE210 HW10 Boyas

1) $CaC_{2(s)} + 5H_2O_{(L)} \rightarrow CaO_{(s)} + 2CO_{2\,(g)} + 5H_{2\,(g)}$

$\Delta\hat{H}_r^\circ = +69.36\ kJ/mol$

(a) The reaction is endothermic because it is absorbing heat. You would need to heat the the reactor to keep the temperature constant. If it ran adiabatically, then the temperature would drop and cool up. More energy is required to break the bonds from the reactants than to form the product bonds ✓

$\frac{20}{20}$

1) 20
2) 16
3) 20
4) 18
5) 0

$\frac{74}{100}$

(b) $\Delta U_r^\circ ?$ $\Delta U_r^\circ(T) = \Delta\hat{H}_r(T) - RT\left[\sum_{\text{Products}} V_i - \sum_{\text{Reactants}} V_i\right]$ Gases

$\left[\sum_p V_i - \sum_r V_i\right] \rightarrow (1+2+5) - 0 = 7$

$\Delta U_r^\circ[T] = +69.36\ \frac{kJ}{mol} - 8.314\ \frac{J}{mol\cdot\cancel{K}} \times (25^\circ C + 273)\cancel{K} \times (7)$

$\Delta U_r^\circ(T) = 69.36 - .0083\ \frac{kJ}{mol}(298)(7)$

$\Delta U_r^\circ(T) = \boxed{52.017\ \frac{kJ}{mol}}$ [amount of energy from the reactants] → Internal energy ✓

(c) $\Delta U = Q + \cancel{W}$ $\Delta U = Q$ $\Delta\hat{U}_r\, n_{CaC_2} = Q$ $m = 150\ g$ $M = 64.1\ g/mol$

@ constant V

$n_{CaC_2} = \frac{m}{M} = 2.34\ mol$ $2.34\ \cancel{mol}\left(52.017\ \frac{kJ}{\cancel{mol}}\right) = \boxed{Q = 121.72\ kJ}$ Heat out of the reactor ✓

Murphy 6.32

$\frac{16}{20}$

$U_{sys,f} - U_{sys,o} = Q$ For Water $C_p = C_v = 4.18\ \frac{J}{g^\circ C}$

$U_{sys,f} - U_{sys,o} = Q = mC_v(T_f - T_o) \rightarrow 650\ gW \times 4.18\ \frac{J}{gW^\circ C}(36.2 - 23.6)^\circ C$

$34234.2\ J \cong \boxed{34234\ J}$ ✓

$\Delta\hat{H}_r = \Delta\hat{U}_r + R(T)$ $n_{Reactant} = \frac{m}{M}$ $\frac{2.7\ g}{154\ g/gmol} = .0175\ gmol$ ✓

$\Delta\hat{H}_r = 1956.23\ \frac{kJ}{gmol} + .008314\ \frac{kJ}{gmol\,\cancel{K}}(298)\cancel{K}$ $\Delta\hat{U}_r n_r = Q$ $\frac{34.234\ kJ}{.0175\ gmol} = 1956.23$ $\Delta\hat{U}_r$

$\boxed{\Delta\hat{H}_r = -1958.7\ \frac{kJ}{gmol}}$ ✓ → $\Delta H_{f\,C_8H_{10}O_3}^\circ = ?$

Murphy 6.34)

$$2Fe + \frac{3}{2}O_2 \rightarrow F_2O_3 \qquad \Delta\hat{H}_f = -830.5\ kJ/gmol$$

20/20

Q = -600 kJ desired (-100 kJ/k × 6k) [released (-)]

$$\frac{-600\ kJ}{-830.5\ kJ/gmol} = .722\ gmol\ Fe_2O_3 \times \frac{2\ gmol\ Fe}{1\ gmol\ Fe_2O_3} = 1.44\ gmol\ Fe$$

$$1.44\ gmol\ Fe \times \frac{56\ g}{1\ gmol} = \boxed{80.64\ g\ Fe}$$

Oxygen $.722\ mol\ Fe_2O_3 \times \frac{160\ g\ Fe_2O_3}{1\ gmol\ Fe_2O_3} = 115\ g\ Fe_2O_3$

am $115g - 80g = 35g$ % $\frac{115-80}{80} \times 100 = 43.75\%$

2)

$$\frac{1}{2}N_{2(g)} + O_2 \rightarrow NO_2$$

8/20

(a)

$$\Delta H_r^0 = H^0_{f\,NO_2} - \frac{1}{2}\cancel{H^0_{f\,N_2}} + \cancel{H^0_{f\,O_2}} \qquad \boxed{\Delta H_r^0 = 33.8\ kJ/mol}$$

180.74

(b) $C_4H_{10} + \frac{13}{2}O_{2(g)} \rightarrow 4CO_{2(g)} + 5H_2O_{(g)}$

$$\Delta H_r = 4H_{f\,CO_2} + 5H_{f\,H_2O} - H_{f\,C_4H_{10}} - \frac{13}{2}\cancel{H_{f\,O_2}}$$

$$\Delta\hat{H}_r = 4 \times (393.5) + 5(-242) - (-125) = \boxed{\Delta\hat{H}_r = 489\ kJ/mol}$$

-2121.2

(c) $C_7H_{16} + 11O_2 \rightarrow 7CO_2 + 8H_2O$

$H_{f\,O_2} = 0$

$$\Delta\hat{H}_r = 7(393.5) + 8(-242) - (-225.2) \qquad \boxed{\Delta\hat{H}_r = 693.7\ kJ/mol}$$

-3855

(d)

$$\overset{(1)}{NaSO_4} + \overset{(2)}{CO} \rightarrow \overset{(3)}{NaSO_3} + \overset{(4)}{CO_2}$$

$$\Delta H_r = (3) + (4) - (2) - (1)$$

$$(298.15) + (393.5) - (-99) - (-1387.1) \qquad \boxed{\Delta H_r = 2177.75\ kJ/mol}$$

-138.2

5)

0/20